

[25] Index for rating diagnostic tests

Authors: W. J. Youden

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Executive Academic Summary & Connectomic Relevance

Formulates Youden's J index ($J = \text{Sensitivity} + \text{Specificity} - 1$) for optimal classification threshold selection in medical diagnostic testing. Used in our Notebook 07 and Notebook 12 to find the optimal decision boundary.

Mathematical & Neurobiological Foundations

- $J = \text{Sensitivity} + \text{Specificity} - 1 = \text{TPR} - \text{FPR}$
- $\theta^* = \arg\max_{\theta} J(\theta)$

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